A Novel Composite Observer Based Approach for Dynamic Event-Triggered Adaptive Fuzzy Control of Nonlinear Systems Under DoS Attacks

Xinjun Wang, Shenghang Liu, Ben Niu, Xinmin Song, Huanqing Wang, and Xudong Zhao

***Abstract*—In this article, we propose a dynamic event-triggered adaptive prescribed-time output feedback tracking control strategy for nonlinear systems with unknown external disturbances under denial-of-service (DoS) attacks. The presence of malicious intermittent DoS attacks makes the output signal and states of the system unavailable, which in turn leads to more difficulty in the design of the controller. To overcome the above difficulty, we construct a novel composite observer based on the attack compensator, whereby the system states can be reconstructed. Moreover, with the help of the time-varying constraint function, the prescribed-time tracking control problem for nonlinear systems is transformed into the constraint problem of tracking error. At the same time, a new dynamic event-triggered adaptive prescribed-time safety fuzzy controller is built and remains applicable in the systems that operate continuously after the predefined time. The adaptive fuzzy safety control method proposed in this article ensures that the tracking error converges to the user-specified region in the predefined time, all signals of the closed-loop system remain bounded under intermittent DoS attacks, and the repeated transmission of samples from the controller to the actuator can be further reduced, generating fewer events and saving communication resources. Finally, the simulation results of the single-link robotic arm demonstrate the rationality and effectiveness of the developed control algorithm.**

***Index Terms*—Compensator-based composite observer, denial-of-service (DoS) attacks, dynamic event triggering, prescribed-time fuzzy tracking control, uncertain nonlinear systems.**

I. Introduction

In RECENT years, with the deep application of the network in all areas of the world, the networked control systems (NCSs) have become more and more popular due to their flexible structure, intelligence, and expandability [1], [2]. For NCSs, they enable communication resources to be shared among various system components and effectively break through the limitations of physical space. However, the NCSs are vulnerable to malicious network attacks from adversaries [3]. For example, replay attacks [4], deception attacks [5], [6], [7], [8], denial-of-service (DoS) [9], [10], [11] attacks. Among them, DoS attacks are a common type of attack that poses nonnegligible risks to productive life, which in turn has received extensive attention from scholars. The problem of adaptive security control for a class of nonlinear systems under DoS attacks was investigated in [9]. The issue of secure control for switched nonlinear systems under DoS attacks was addressed by [12]. For a class of nonlinear multiagent systems (MASs), reference [13] was the first to study the problem of cooperative output regulation under DoS attacks and proposed a resilient distributed control scheme. Since the system to resist network attacks needs to consume more resources, how to reasonably and effectively save the limited system resources is a hot issue in current research.

As the NCSs continue to develop, more data needs to be transmitted continuously, which leads to the existing communication resources unaffordable. Fortunately, a feasible way to alleviate the load of network data transmission is to employ the event-triggered mechanism (ETM) [14]. For nonlinear continuous-time systems, an event-triggered integral sliding mode control strategy was proposed in [15]. An adaptive event-triggered mechanism for Takagi–Sugeno fuzzy systems with multiple cyber-attacks was explored in [16]. However, the above literatures discuss the static ETM (SETM). In order to further improve resource utilization, the authors in [17] dynamically adjusted the threshold parameters with the help of a positive auxiliary variable, which in turn proposed the dynamic ETM (DETM). The DETM can not only maintain similar performance as the SETM, but also further reduce the frequency of events. For linear MASs, a decentralized dynamic event-triggered control mechanism was proposed in [18]. For nonlinear systems with unknown growth rates [19], a dynamic event-triggered scheme based on the dynamic gain technique was developed. The dynamic event-triggered adaptive tracking control problem for nonlinear MASs with time-varying delays was investigated in [20]. It is worth noting that the effects of cyber-attacks are not considered in the above research contents, and how to realize the dynamic event-triggered

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tracking control strategy under cyber-attacks is a challenging topic.

In certain areas that rely on the NCSs, such as space docking, missile interception, and drone formations, achieving ideal tracking accuracy within a short timeframe is essential. In such cases, the prescribed-time control scheme plays a crucial role. For linear systems, a control strategy for full-state regulation within the prescribed-time was established in [21]. For a class of strict-feedback nonlinear systems, a new practical predefined time stable criterion was first presented in [22]. The predefined-time adaptive fuzzy bipartite consensus control problem for multiquadrotors under malicious attacks was studied in [23]. Based on the definition of the general prescribed-time function, it is not difficult to see that as the time t approaches the predefined time T, it may lead to the issue of infinite gain. In order to conquer this obstacle, a control strategy using partial tracking error constraints based on the performance function for nonlinear systems was proposed in [24]. The above literatures generally assume that the initial set satisfies $-g(0) \leq z_1(0) \leq g(0)$, in which $g(t)$ is the prescribed-time constraint function, as well as $z_1(0)$ is the tracking error of the system. However, this approach cannot be applied when the system's preliminary values are infinite, which urgently demands that we consider more general assumptions.

Motivated by the above discussions, we focus on solving the dynamic event-triggered adaptive prescribed-time security fuzzy tracking control problem of uncertain nonlinear systems under DoS attacks. The main contributions of this article are as follows.

1) When the enemy carries out a malicious DoS attack, the output signal of the system is not available at this time, which renders the conventional fuzzy observer designed in [25] and [26] unable to work properly. Different from [9] and [10], we design a new composite fuzzy observer by introducing an attack compensator and using the compensated output signal, which not only reconstructs the output signal y as well as the states of the system, but also studies the unknow external disturbances.
2) With the help of the time-varying constraint function, a new adaptive prescribed-time fuzzy controller is designed, and the controller can be applied to the system that operates continuously after the predefined settling time, circumventing the infinite gain phenomenon in the literatures [27] and [28]. The predefined time and tracking accuracy involved in this article are independent of the initial conditions and other design parameters, which are only set on demand by the user according to the actual engineering context. In addition, for the assumption about the initial value of the system in [24] and [29], we extend it to the more general assumption that the initial values can be infinite.
3) Unlike the SETMs proposed in [14], [15], and [16], a DETM is designed in this article by introducing an auxiliary dynamic variable, which reduces repeated data transmission and thus saves communication resources and improves transmission efficiency. In addition, the command filtering technique is considered to avoid repetitive differentiation of the virtual control signals in the backstepping design, and the fuzzy logic systems (FLSs) are utilized to approximate the unknown nonlinear functions.

II. Mathematical Preliminaries and Problem Formulation

A. System Descriptions

The following uncertain nonlinear system is considered in this article:

$$\begin{cases} \dot{x}_i = x_{i+1} + f_i(\bar{x}_i) + \epsilon_i(t), \\ \dot{x}_n = u + f_n(\bar{x}_n) + \epsilon_n(t), \\ y = x_1, \quad i = 1, \ldots, n-1 \end{cases} \tag{1}$$

where $\bar{x}_i = [x_1, \ldots, x_i]^T \in \Re^i$ is the state vector of the system with $i = 1, \ldots, n$, $u \in \Re$ is the designed controller and $y \in \Re$ is the output of the system. In addition, $f_i(\star)$ and $\epsilon_i(t)$ are the unknown smooth nonlinear function and unknown external disturbance, respectively.

Assumption 1: The reference signal y_r is n-order differentiable.

Assumption 2: The continuous time-varying disturbance $\epsilon_i(t)$ is bounded and satisfy $|\epsilon_i(t)| \leq \epsilon_{Mi}$ with ϵ_{Mi} being positive constant.

Remark 1: For the nonlinear systems tracking problem, Assumption 1 is not only a fundamental preliminary assumption for the study of this problem, but also a prerequisite for the subsequent control design in this article. This assumption can be found in [30] and [31]. In many real physical systems, external disturbances are often bounded by natural or engineering constraints with a certain range. Thus, Assumption 2 reflects actual physical constraint and is a reasonable engineering assumption from the system design perspective. Moreover, this assumption can be found in [32] and [33].

B. Fuzzy Logic Systems (FLSs)

In this article, the FLSs are used to approximate an unknown continuous nonlinear function $f(x)$ that is defined on some compact set. The property of FLSs can be further expressed using the following lemma:

Lemma 1 (See [11]): For any continuous function $f(x)$ which is defined on the compact set $\mathcal{K}$, there exists a fuzzy logic system such that

$$\sup_{x \in \mathcal{K}} \left| f(x) - \omega^T \mathcal{S}(x) \right| \leq H \tag{2}$$

where H is a given positive constant. $\omega(x) = [\omega_1(x), \ldots, \omega_M(x)]^T$ is defined as the ideal constant vector and M stands the number of the fuzzy rules with $M > 1$. In addition, $\mathcal{S}(x) = [\mathcal{S}_1(x), \ldots, \mathcal{S}_M(x)]^T / \sum_{i=1}^{M} \mathcal{S}_i(x)$ denotes the fuzzy basis function vector and the following property is satisfied $0 < \mathcal{S}^T(x)\mathcal{S}(x) \leq 1$. Besides, $\mathcal{S}_i(x)$ $(i = 1, \ldots, M)$

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can be constructed as the following Gaussian function:

$$\mathcal{S}_i(x) = \exp\left(\frac{-(x-\mu_i)^T(x-\mu_i)}{D_i^2}\right) \tag{3}$$

where $\mu_i = [\mu_{i,1}, \ldots, \mu_{i,n}]^T$ represents the center vector and D_i is the width of the Gaussian function.

Combined with the FLSs, the following equation holds:

$$f_k(\bar{x}_k) = \omega_k^T \mathcal{S}(\bar{x}_k) + \chi_k(\bar{x}_k), k = 1, \ldots, n \tag{4}$$

where $\chi_k(\bar{x}_k)$ is the estimation error function with $|\chi_k(\bar{x}_k)| \leq \bar{\chi}_k$ and ω_k being the ideal weight vector.

C. Intermittent DoS Attacks

In this article, we consider a highly representative type of cyber-attack, which is the intermittent DoS attacks. This means that the attacks are intermittent rather than continuous. Let T_l^d and T_l^s represent the start and end instant of the lth DoS attack, respectively, and thus we can get the lth DoS attack interval $T_l := [T_l^d, T_l^s)$. Based on the above, it is easy to know that during this time interval, the DoS attack is active and data cannot be transmitted in the communication channels.

For this purpose, we define the time sum of DoS attacks occurring in the time interval $[\varepsilon, t]$ as the following form:

$$\Psi_d(\varepsilon, t) = \bigcup_{l \in N} T_l \bigcap [\varepsilon, t] \tag{5}$$

where $N > 0$ is an integer.

Then, the sum of attack sleep times are defined as

$$\Psi_s(\varepsilon, t) = [\varepsilon, t] \setminus \Psi_d(\varepsilon, t). \tag{6}$$

Assumption 3 (DoS Duration [10]): There exists a constant $\mathcal{T} \geq 1$ and satisfies

$$|\Psi_d(\varepsilon, t)| \leq \frac{t-\varepsilon}{\mathcal{T}}. \tag{7}$$

Assumption 4 (DoS Frequency [10]): There exist constants $\eta > 0$ and $\tau_D > 0$ satisfying

$$n(\varepsilon, t) \leq \eta + \frac{t-\varepsilon}{\tau_D}. \tag{8}$$

Remark 2: In practical engineering applications, due to limited resources, attackers often prefer to launch multiple attacks in a short period of time, which means that the attack methods used are nonperiodic rather than periodic. In order to be more in line with the practical needs of users, this article focuses on how to counteract nonperiodic DoS attacks. Therefore, the restrictions on the duration and frequency of DoS attacks regarding Assumptions 3 and 4 are reasonable, and the above assumptions can be found in [10] and [34].

D. Dynamic Event-Triggered Mechanism

In order to reduce unnecessary signal transmission in NCSs and save communication resources, the following DETM is designed:

$$\begin{aligned} u(t) &= u^*(t_k), t \in [t_k, t_{k+1}), k \in \mathbb{N} \\ t_{k+1} &= \inf\{t > t_k | \gamma + o(\hbar - |e_u|) \leq 0\} \end{aligned} \tag{9}$$

where u is the actual control signal that applied to the system and u^* the final control signal. γ is defined as an external auxiliary variable that can be obtained from the following dynamic variable:

$$\dot{\gamma} = -r_u \gamma + o(\hbar - |e_u|) \tag{10}$$

with $\hbar$, o, r_u being the design positive constants, $e_u = u - u^*$. It can be obtained from (9) and (10) that we can further save communication resources by dynamically adjusting threshold parameters with the help of auxiliary variable γ. In addition, based on the above DETM, there exists a constant Δ such that $|u - u^*| \leq \Delta$ holds all the time.

Lemma 2 (See [35]): Considering the external auxiliary variable (10) and DETM (9), the variable γ satisfies $0 \leq \gamma \leq \frac{o\hbar}{r_u}$ for $0 \leq t$.

Considering the uncertain nonlinear system (1) under DoS attacks, the main control objective of this work is to design a safety fuzzy controller based on composite observer under Assumptions 1–4 to achieve the following goals.

1) The output y of system tracks the reference trajectory y_r in the predefined time T.
2) All signals involved in the closed-loop system are guaranteed to be bounded.
3) The Zeno behavior caused by DETM is avoided.

III. Main Results

The main work of this section will be divided into the following three parts: 1) a composite state observer based on the attack compensator will be built to reconstruct the unmeasurable state variables; 2) based on the composite observer, a dynamic event-triggered adaptive prescirbed-time safety fuzzy control strategy is proposed; and 3) the stability of the system will be effectively demonstrated.

A. Compensator-Based Composite Observer Design

Let $\check{y}$ denote the actual measurement output when system (1) is attacked, the attack compensator can be designed as

$$\check{y} = \begin{cases} x_1(T_l^{d-}), & t \in [T_l^d, T_l^s), \\ x_1(t), & t \in [T_l^s, T_{l+1}^d) \end{cases} \tag{11}$$

where T_l^{d-} is the left limit of T_l^d. Define $y^d = x_1(T_l^{d-})$ and $y^s = x_1(t)$ with $x_1(T_l^{d-})$ being the last obtained output signal before the lth DoS attack occurs.

Considering the reconstruction of unmeasurable system states x_i under DoS attacks, a novel compensator-based composite observer is designed

$$\begin{cases} \dot{\hat{x}}_i = \hat{x}_{i+1} + \hat{\omega}_i^T \mathcal{S}_i(\hat{\bar{x}}_i) + \hat{\epsilon}_i - \ell_i(\check{y} - \hat{y}), \\ \dot{\hat{x}}_n = u + \hat{\omega}_n^T \mathcal{S}_n(\hat{\bar{x}}_n) + \hat{\epsilon}_n - \ell_n(\check{y} - \hat{y}), \\ \hat{y} = \hat{x}_1, \quad i = 1, \ldots, n-1 \end{cases} \tag{12}$$

where $\hat{x}_i (i = 1, \ldots, n)$ is the estimate of x_i, $\hat{\bar{x}}_i = [\hat{x}_1, \ldots, \hat{x}_n]^T$. ℓ_i is the designed observer gain. Besides, $\hat{\omega}_i$ is the estimate of ω_i. When DoS attacks occur, $\hat{\omega}_i$ is restated as $\hat{\omega}_{di}$; otherwise, $\hat{\omega}_i$ is represented as $\hat{\omega}_{si}$.

Let $e_i = x_i - \hat{x}_i$ be an estimate error with $i = 1, \ldots, n$. Combining (1) and (12), the error e is further satisfied in the following

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form:

$$\dot{e}=\begin{cases}\mathcal{A}'e+\mathcal{L}_dMe_1^d+\Omega_d(\hat{\bar{x}}_n)\tilde{\mathcal{W}}_d+\Phi_d+\tilde{\mathcal{E}}_d+\chi_d(\bar{x}_n),\\ t\in[T_l^d,T_l^s),\\ \mathcal{A}e+\Omega_s(\hat{\bar{x}}_n)\tilde{\mathcal{W}}_s+\Phi_s+\tilde{\mathcal{E}}_s+\chi_s(\bar{x}_n),\\ t\in[T_l^s,T_{l+1}^d)\end{cases}\tag{13}$$

where $e=[e_1,e_2,\ldots,e_n]^T$ and we define $e_1^d=y^d-\hat{y}$. $\mathcal{A}=\mathcal{A}'+\mathcal{L}_sM$ and $\mathcal{A}'=\begin{pmatrix}0 & I_{n-1}\\ 0 & 0\end{pmatrix}$ are both Hurwitz matrices with $M=[1,0,\ldots,0]$. $\mathcal{L}_d=[\ell_1^d,\ldots,\ell_n^d]^T$, $\mathcal{L}_s=[\ell_1^s,\ldots,\ell_n^s]^T$ and $\tilde{\mathcal{E}}_d=[\tilde{\epsilon}_{d1},\ldots,\tilde{\epsilon}_{dn}]^T,\tilde{\mathcal{E}}_s=[\tilde{\epsilon}_{s1},\ldots,\tilde{\epsilon}_{sn}]^T$,

$$\begin{aligned}&\tilde{\mathcal{W}}_d=[\tilde{\omega}_{d1}^T,\ldots,\tilde{\omega}_{dn}^T]^T,\tilde{\mathcal{W}}_s=[\tilde{\omega}_{s1}^T,\ldots,\tilde{\omega}_{sn}^T]^T,\\ &\chi_d=[\chi_{d1}(x_1),\ldots,\chi_{dn}(\bar{x}_n)]^T,\\ &\chi_s=[\chi_{s1}(x_1),\ldots,\chi_{sn}(\bar{x}_n)]^T,\\ &\Omega_d=diag\{S_1^d(\hat{x}_1),\ldots,S_n^d(\hat{\bar{x}}_n)\},\\ &\Omega_s=diag\{S_1^s(\hat{x}_1),\ldots,S_n^s(\hat{\bar{x}}_n)\},\\ &\Phi_d=[\omega_{d1}^T(S_1^d(x_1)-S_1^d(\hat{x}_1)),\ldots,\omega_{dn}^T(S_n^d(\bar{x}_n)-S_n^d(\hat{\bar{x}}_n))]^T,\\ &\Phi_s=[\omega_{s1}^T(S_1^s(x_1)-S_1^s(\hat{x}_1)),\ldots,\omega_{sn}^T(S_n^s(\bar{x}_n)-S_n^s(\hat{\bar{x}}_n))]^T\end{aligned}\tag{14}$$

with $\tilde{\omega}_{di}^T=\omega_{di}^T-\hat{\omega}_{di}^T$ and $\tilde{\omega}_{si}^T=\omega_{si}^T-\hat{\omega}_{si}^T$.

In addition, we construct a disturbance observer to model the unmeasurable disturbances. First, an intermediate variable is introduced

$$\lambda_i=\epsilon_i-\vartheta_ix_i,i=1,\ldots,n.\tag{15}$$

When the system is under the DoS attacks, λ_i is denoted as λ_{di}; conversely, it is restated as λ_{si}. Then, we can get the differentiation of λ_i as

$$\begin{aligned}\dot{\lambda}_i&=\dot{\epsilon}_i-\vartheta_i\left(x_{i+1}+\omega_i^T\mathcal{S}_i(\bar{x}_i)+\chi_i(\bar{x}_i)+\epsilon_i+\vartheta_ix_i\right),\\ \dot{\lambda}_n&=\dot{\epsilon}_n-\vartheta_n\left(u+\omega_n^T\mathcal{S}_n(\bar{x}_n)+\chi_n(\bar{x}_n)+\epsilon_n+\vartheta_nx_n\right)\end{aligned}\tag{16}$$

where ϑ_i is the designed positive number.

For more convenient subsequent design, the estimation of $\dot{\hat{\lambda}}_i$ can be expressed as follows:

$$\begin{aligned}\dot{\hat{\lambda}}_i&=-\vartheta_i(\hat{x}_{i+1}+\hat{\omega}_i^T\mathcal{S}_i(\hat{\bar{x}}_i)+\hat{\epsilon}_i+\vartheta_i\hat{\bar{x}}_i),\\ \dot{\hat{\lambda}}_n&=-\vartheta_n(u+\hat{\omega}_n^T\mathcal{S}_n(\hat{\bar{x}}_n)+\hat{\epsilon}_n+\vartheta_n\hat{\bar{x}}_n).\end{aligned}\tag{17}$$

Combined with the above analysis, $\hat{\epsilon}_i$ can be further expressed as $\hat{\epsilon}_i=\hat{\lambda}_i+\vartheta_i\hat{x}_i$. Let $\tilde{\lambda}_i=\lambda_i-\hat{\lambda}_i$. From (16) and (17), it is easy to obtain that the following equations hold true:

$$\begin{aligned}\dot{\tilde{\lambda}}_i&=\dot{\tilde{\epsilon}}_i-\vartheta_i[e_{i+1}+\tilde{\lambda}_i+\vartheta_ie_i+\chi_i(\bar{x}_i)+\tilde{\omega}_i^T\mathcal{S}_i(\hat{\bar{x}}_i)\\ &\quad+\omega_i^T(\mathcal{S}_i(\bar{x}_i)-\mathcal{S}_i(\hat{\bar{x}}_i))],\\ \dot{\tilde{\lambda}}_n&=\dot{\tilde{\epsilon}}_i-\vartheta_n[\tilde{\lambda}_n+\vartheta_ne_n+\chi_n(\bar{x}_n)+\tilde{\omega}_n^T\mathcal{S}_n(\hat{\bar{x}}_n)\\ &\quad+\omega_n^T(\mathcal{S}_n(\bar{x}_n)-\mathcal{S}_n(\hat{\bar{x}}_n))].\end{aligned}\tag{18}$$

For $t\in[T_l^d,T_l^s)$, we can express the definition of e_1^d as follows: $e_1^d=y^d-\hat{y}$. Then, the Lyapunov function can be selected as $\mathcal{V}_0^d=e^T\mathcal{P}_de+\frac{1}{2}\sum_{i=1}^n\tilde{\lambda}_{di}^2$, where $\mathcal{P}_d>0$ is a symmetric matrix to be given later. The derivative of $\mathcal{V}_0^d$ is given as

$$\begin{aligned}\dot{\mathcal{V}}_0^d&=e^T(\mathcal{P}_d\mathcal{A}'+\mathcal{A}'^T\mathcal{P}_d)e+2e^T\mathcal{P}_d[\mathcal{L}_dM_de_1^d+\tilde{\mathcal{E}}_d\\ &\quad+\Omega_d(\hat{\bar{x}}_n)\tilde{\mathcal{W}}_d+\Phi_d+\chi_d(\bar{x}_n)]+\sum_{i=1}^n\tilde{\lambda}_{di}\left[\dot{\epsilon}_{di}\right.\\ &\quad-\vartheta_{di}\left(e_{i+1}+\tilde{\lambda}_{di}+\vartheta_{di}e_i+\chi_{di}(\bar{x}_i)+\tilde{\omega}_{di}^T\mathcal{S}_i^d(\hat{\bar{x}}_i)\right.\\ &\quad\left.\left.+\omega_{di}^T(\mathcal{S}_i^d(\bar{x}_i)-\mathcal{S}_i^d(\hat{\bar{x}}_i))\right)\right].\end{aligned}\tag{19}$$

Based on Young's inequality and the propertiy of FLSs, the following inequalities hold:

$$\begin{aligned}&2e^T\mathcal{P}_d\mathcal{L}_dM_de_1^d\le\left(\frac{1}{k}+k\|\mathcal{P}_d\mathcal{L}_dM_d\|^2\right)\|e\|^2,\\ &2e^T\mathcal{P}_d\Omega_d(\hat{\bar{x}}_n)\tilde{\mathcal{W}}_d\le e^T\mathcal{P}_d^T\mathcal{P}_de+\sum_{i=1}^n\tilde{\omega}_{di}^T\tilde{\omega}_{di},\\ &2e^T\mathcal{P}_d[\Phi_d+\chi_d(\bar{x}_n)]\le3e^T\mathcal{P}_d^T\mathcal{P}_de+||\bar{\chi}_d||^2+2||\mathcal{W}_d||^2,\\ &2e^T\mathcal{P}_d\tilde{\mathcal{E}}_d\le2e^T\mathcal{P}_d^T\mathcal{P}_de+\vartheta_{dM}^2||e||^2+\sum_{i=1}^n\tilde{\lambda}_{di}^2,\\ &\sum_{i=1}^n\tilde{\lambda}_{di}\dot{\epsilon}_{di}\le\frac{1}{2}\sum_{i=1}^n\tilde{\lambda}_{di}^2+\frac{1}{2}\sum_{i=1}^n\epsilon_{Mi}^2,\\ &\sum_{i=1}^n\tilde{\lambda}_{di}(-\vartheta_{di}e_{i+1}-\vartheta_{di}^2e_i)\le\frac{\vartheta_{dM}^2+\vartheta_{dM}^4}{2}||e||^2+\sum_{i=1}^n\tilde{\lambda}_{di}^2,\\ &-\sum_{i=1}^n\tilde{\lambda}_{di}\vartheta_{di}(\chi_{di}(\bar{x}_i)+\omega_{di}^T(\mathcal{S}_i^d(\bar{x}_i)-\mathcal{S}_i^d(\hat{\bar{x}}_i))+\tilde{\omega}_{di}^T\mathcal{S}_i^d(\hat{\bar{x}}_i))\\ &\le\sum_{i=1}^n\vartheta_{dM}^2\tilde{\lambda}_{di}^2+\sum_{i=1}^n\tilde{\omega}_{di}^T\tilde{\omega}_{di}+||\bar{\chi}_d||^2+2||\mathcal{W}_d||^2\end{aligned}\tag{20}$$

where $\vartheta_{dM}=\max\{\vartheta_{di},i=1,\ldots,n\}$, k is a positive constant. Moreover, $\mathcal{W}_d=[\omega_{d1}^T,\ldots,\omega_{dn}^T]^T$ and $\bar{\chi}_d=[\bar{\chi}_{d1},\ldots,\bar{\chi}_{dn}]^T$.

Combining (19) and (20), we can obtain the following inequality:

$$\begin{aligned}\dot{\mathcal{V}}_0^d\le\ &e^T\left(\mathcal{P}_d\mathcal{A}'+\mathcal{A}'^T\mathcal{P}_d+6\mathcal{P}_d^T\mathcal{P}_d+\left(\frac{1}{k}+k\|\mathcal{P}_d\mathcal{L}_dM_d\|^2\right.\right.\\ &\left.\left.+\frac{3\vartheta_{dM}^2+\vartheta_{dM}^4}{2}\right)I_n\right)e+2\sum_{i=1}^n\tilde{\omega}_{di}^T\tilde{\omega}_{di}+q_{d0}\\ &-\sum_{i=1}^n\left(\vartheta_{di}-\frac{5}{2}-\vartheta_{dM}^2\right)\tilde{\lambda}_{di}^2\end{aligned}\tag{21}$$

where $q_{d0}=\frac{1}{2}\sum_{i=1}^n\epsilon_{Mi}^2+2\|\bar{\chi}_d\|^2+4\|\mathcal{W}_d\|^2$ and I_n is the unit matrix.

When $t\in[T_l^s,T_{l+1}^d)$, we have that $e_1^s=y^s-\hat{y}$. The following Lyapunov function is chosen: $\mathcal{V}_0^s=e^T\mathcal{P}_se+\frac{1}{2}\sum_{i=1}^n\tilde{\lambda}_{si}^2$, where $\mathcal{P}_s>0$ is symmetric matrix, as will be shown below. The derivative of $\mathcal{V}_0^s$ is given as

$$\begin{aligned}\dot{\mathcal{V}}_0^s=\ &e^T(\mathcal{P}_s\mathcal{A}+\mathcal{A}^T\mathcal{P}_s)e+2e^T\mathcal{P}_s[\tilde{\mathcal{E}}_s+\Omega_s(\hat{\bar{x}}_n)\tilde{\mathcal{W}}_s\\ &+\Phi_s+\chi_s(\bar{x}_n)]+\sum_{i=1}^n\tilde{\lambda}_{si}\left[\dot{\epsilon}_{si}-\vartheta_{si}\left(e_{i+1}+\tilde{\lambda}_{si}\right.\right.\end{aligned}$$

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$$+\vartheta_{si}e_i+\chi_{si}(\bar{x}_i)+\tilde{\omega}_{si}^T\mathcal{S}_i^s(\hat{\bar{x}}_i)+\omega_{si}^T(\mathcal{S}_i^s(\bar{x}_i)-\mathcal{S}_i^s(\hat{\bar{x}}_i))\big)\big]. \tag{22}$$

Similar to the treatment of (20), (22) can be further transformed into the following inequality:

$$\dot{\mathcal{V}}_0^s \le e^T\left(\mathcal{P}_s\mathcal{A}+\mathcal{A}^T\mathcal{P}_s+6\mathcal{P}_s^T\mathcal{P}_s+\frac{3\vartheta_{sM}^2+\vartheta_{sM}^4}{2}I_n\right)e + 2\sum_{i=1}^{n}\tilde{\omega}_{si}^T\tilde{\omega}_{si}+q_{s0}-\sum_{i=1}^{n}\left(\vartheta_{si}-\frac{5}{2}-\vartheta_{sM}^2\right)\tilde{\lambda}_{si}^2 \tag{23}$$

where $q_{s0}=\frac{1}{2}\sum_{i=1}^{n}\epsilon_{Mi}^2+2\|\bar{\chi}_s\|^2+4\|\mathcal{W}_s\|^2$ and I_n is the unit matrix. Besides, $\vartheta_{sM}=\max\{\vartheta_{si},i=1,\ldots,n\}$, $\mathcal{W}_s=[\omega_{s1}^T,\ldots,\omega_{sn}^T]^T$ and $\bar{\chi}_d=[\bar{\chi}_{s1},\ldots,\bar{\chi}_{sn}]^T$.

B. Adaptive Prescribed-Time Security Controller Design

In this article, we define the tracking error as

$$z_1=\tilde{y}-y_r \tag{24}$$

where

$$\tilde{y}=\begin{cases}\hat{x}_1, & t\in[T_l^d,T_l^s),\\ x_1, & t\in[T_l^s,T_{l+1}^d).\end{cases} \tag{25}$$

Combining the predefined time $\mathcal{T}\in(0,\infty)$ and the pregiven tracking accuracy $b\in(0,\infty)$, the following time-varying constraint function can be constructed:

$$g(t)=\begin{cases}b, & t>\mathcal{T},\\ (\frac{1}{t}-\frac{1}{\mathcal{T}})^{2\flat}+b, & 0\le t\le\mathcal{T}\end{cases} \tag{26}$$

where $\flat$ is a positive integer satisfying $2\flat>n+1$ with n representing the order of the system. This is to ensure that the function $g(t)$ is continuously differentiable.

Then, in order to achieve the desired prescribed-time tracking objective, the following tracking error constraint need to be satisfied:

$$-g(t)\le z_1(t)\le g(t). \tag{27}$$

Combined with the definition of the scalar function g given in (26), we can obtain from (27) that

$$|z_1(t)|<b,\quad t\ge\mathcal{T}. \tag{28}$$

From the above analysis, we learn that the tracking error z_1 converges to a specified range b within the predefined time $\mathcal{T}$, which can be achieved by ensuring (27).

Since the constraint boundary in (26) is infinite with respect to the initial time of the system, the following constraint control strategy is proposed to solve this problem:

$$\psi(\varrho(t))=\begin{cases}-(\frac{\varrho(t)}{\iota}-1)^{2\flat}+1, & 0<\varrho(t)\le\iota,\\ 1, & \varrho(t)>\iota\end{cases} \tag{29}$$

where $\varrho(t)=a(g(t)-z_1(t))(g(t)+z_1(t))=a[g^2(t)-z_1^2(t)]$.

In the above, ι and a are positive constants that have been designed. It is worth noting that $\psi(\varrho(t))$ in (29) is the smooth composite function. Moreover, when $\varrho(t)=\iota$, ψ is continuous. Meanwhile, the value of $\varrho(t)$ can be mapped to the interval (0,1] by means of the $\psi(\varrho(t))$.

In this work of controller construction, the following coordinate transformation is designed:

$$\mathfrak{z}_1(t)=\frac{z_1(t)}{\psi(\varrho(t))}=\frac{z_1(t)}{\psi(a(g^2(t)-z_1^2(t)))}. \tag{30}$$

Lemma 3 (See [36]): If $\mathfrak{z}_1\in\mathbb{L}_\infty$, then $-g(t)<z_1<g(t)$ always holds for $\forall t>0$, else $z_1\to g(t)$ or $z_1\to -g(t)$.

According to the observer (12), (30) and the first-order filter, the following coordinate transformations are designed:

$$\hbar_1=\mathfrak{z}_1,\quad \hbar_j=\hat{x}_j-\nu_j, j=2,\ldots,n \tag{31}$$

where ν_j is the output of the following command filter. It is related to the virtual controller α_{j-1}

$$\varpi_j\dot{\nu}_j+\nu_j=\alpha_{j-1},\nu_j(0)=\alpha_{j-1}(0) \tag{32}$$

where ϖ_j is the positive design constant. In addition, we define the filter error as $\varsigma_j=\nu_j-\alpha_{j-1}$.

Next, we will divide the backoff process into different time intervals: $t\in[T_l^d,T_l^s)$ and $t\in[T_l^s,T_{l+1}^d)$ according to whether the system is under attack or not.

Case 1: When $t\in[T_l^d,T_l^s)$, the system has suffered the DoS attack and is in danger. At this point, the tracking error (24) is $z_{d,1}=\hat{y}-y_r$. Therefore, we have

$$\dot{\hbar}_1^d=\pi_{d1}\left(\hbar_2^d+\varsigma_{d2}+\alpha_{d1}+\omega_{d1}^T\mathcal{S}_1^d(\hat{x}_1)-\tilde{\omega}_{d1}^T\mathcal{S}_1^d(\hat{x}_1)+\hat{\epsilon}_{d1}-\ell_1^d(\check{y}-\hat{x}_1)-\dot{y}_r\right)+\varkappa_{d1} \tag{33}$$

where

$$\pi_{d1}=\begin{cases}\frac{1}{\psi}-\frac{4a\flat}{\iota\psi^2}\left[\frac{\varrho}{\iota}-1\right]^{2\flat-1}z_{d,1}^2>0, & 0<\varrho\le\iota,\\ 1, & \varrho>\iota,\end{cases} \tag{34}$$

$$\varkappa_{d1}=\begin{cases}\frac{4a\flat}{\iota\psi^2}\left[\frac{\varrho}{\iota}-1\right]^{2\flat-1}\dot{g}gz_{d,1}, & 0<\varrho\le\iota,\\ 0, & \varrho>\iota\end{cases}. \tag{35}$$

Step 1: Construct the Lyapunov function $\mathcal{V}_1^d$ as $\mathcal{V}_1^d=\mathcal{V}_0^d+\frac{1}{2}(\hbar_1^d)^2+\frac{1}{2\varphi_{d1}^*}\tilde{\zeta}_{d1}^2+\frac{1}{2\varphi_{d1}}\tilde{\omega}_{d1}^T\tilde{\omega}_{d1}+\frac{1}{2}\varsigma_{d2}^2$, where φ_{d1} and φ_{d1}^* are known positive constants. $\zeta_{d1}=||\omega_{d1}||^2$, $\hat{\zeta}_{d1}$ is the estimate of ζ_{d1} and $\tilde{\zeta}_{d1}=\zeta_{d1}-\hat{\zeta}_{d1}$.

Further, the derivative of $\mathcal{V}_1^d$ is

$$\dot{\mathcal{V}}_1^d=\dot{\mathcal{V}}_0^d+\hbar_1^d\left[\pi_{d1}\left(\hbar_2^d+\varsigma_{d2}+\alpha_{d1}+\omega_{d1}^T\mathcal{S}_1^d(\hat{x}_1)+\hat{\epsilon}_{d1}-\tilde{\omega}_{d1}^T\mathcal{S}_1^d(\hat{x}_1)-\ell_1^d(\check{y}-\hat{x}_1)-\dot{y}_r\right)+\varkappa_{d1}\right]+\frac{1}{\varphi_{d1}}\tilde{\omega}_{d1}^T\dot{\tilde{\omega}}_{d1}+\frac{1}{\varphi_{d1}^*}\tilde{\zeta}_{d1}\dot{\tilde{\zeta}}_{d1}+\varsigma_{d2}\dot{\varsigma}_{d2}. \tag{36}$$

Based on Young's inequality and the property of FLSs, the following inequalities hold true:

$$\begin{aligned}&\hbar_1^d\varkappa_{d1}\le r(\hbar_1^d)^2\varkappa_{d1}^2+\frac{1}{4r},\\ &\hbar_1^d\pi_{d1}\varsigma_{d2}\le\frac{1}{2}\varsigma_{d2}^2+\frac{1}{2}(\hbar_1^d)^2\pi_{d1}^2,\\ &\hbar_1^d\pi_{d1}\hbar_2^d\le\frac{1}{2}(\hbar_1^d)^2\pi_{d1}^2+\frac{1}{2}(\hbar_2^d)^2,\\ &\hbar_1^d\pi_{d1}\omega_{d1}^T\mathcal{S}_1^d(\hat{x}_1)\le\frac{r}{2}(\hbar_1^d)^2\pi_{d1}^2\zeta_{d1}+\frac{1}{2r},\\ &-2\hbar_1^d\pi_{d1}\tilde{\omega}_{d1}^T\mathcal{S}_1^d(\hat{x}_1)\le(\hbar_1^d)^2\pi_{d1}^2+\tilde{\omega}_{d1}^T\tilde{\omega}_{d1}\end{aligned} \tag{37}$$

where $r>0$ is a known constant.

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The following inequality can be obtained by taking (37) into (36):

$$\begin{aligned}\dot{\mathcal{V}}_1^d \leq\ & \dot{\mathcal{V}}_0^d + \hbar_1^d\pi_{d1}\left(2\hbar_1^d\pi_{d1} + \alpha_{d1} + \frac{r}{2}\hbar_1^d\pi_{d1}\hat{\zeta}_{d1} + \hat{\epsilon}_{d1}\right.\\ & \left. -\ell_1^d(\check{y}-\hat{x}_1) - \dot{y}_r + \frac{r}{\pi_{d1}}\hbar_1^d\varkappa_{d1}^2\right) + \tilde{\omega}_{d1}^T\tilde{\omega}_{d1}\\ & + \frac{1}{2}\varsigma_{d2}^2 + \varsigma_{d2}\dot{\varsigma}_{d2} + \tilde{\zeta}_{d1}\left(\frac{r}{2}(\hbar_1^d)^2\pi_{d1}^2 - \frac{1}{\varphi_{d1}^*}\dot{\hat{\zeta}}_{d1}\right)\\ & + \frac{3}{4r} + \tilde{\omega}_{d1}^T\left(\hbar_1^d\pi_{d1}\mathcal{S}_1^d(\hat{x}_1) - \frac{1}{\varphi_{d1}}\dot{\hat{\omega}}_{d1}\right). \end{aligned} \tag{38}$$

The virtual controller α_{d1} is designed as

$$\begin{aligned}\alpha_{d1} =\ & -\frac{\beta_{d1}}{\pi_{d1}}\hbar_1^d - \hat{\epsilon}_{d1} - \frac{r}{\pi_{d1}}\hbar_1^d\varkappa_{d1}^2 - 2\hbar_1^d\pi_{d1}\\ & + \ell_1^d(\check{y}-\hat{x}_1) + \dot{y}_r - \frac{r}{2}\hbar_1^d\pi_{d1}\hat{\zeta}_{d1} \end{aligned} \tag{39}$$

and the adaptive laws are constructed as

$$\begin{aligned}\dot{\hat{\omega}}_{d1} &= \varphi_{d1}\hbar_1^d\pi_{d1}\mathcal{S}_1^d(\hat{x}_1) - p_{d1}\hat{\omega}_{d1}\\ \dot{\hat{\zeta}}_{d1} &= \varphi_{d1}^*\frac{r}{2}(\hbar_1^d)^2\pi_{d1}^2 - k_{d1}\hat{\zeta}_{d1}\end{aligned} \tag{40}$$

where β_{d1}, p_{d1}, and k_{d1} are know positive constants.

Substituting (39) and (40) into (38) results in

$$\begin{aligned}\dot{\mathcal{V}}_1^d \leq\ & \dot{\mathcal{V}}_0^d - \beta_{d1}(\hbar_1^d)^2 + \frac{1}{2}(\hbar_2^d)^2 + \frac{1}{2}\varsigma_{d2}^2 + \varsigma_{d2}\dot{\varsigma}_{d2}\\ & + \tilde{\omega}_{d1}^T\tilde{\omega}_{d1} + \frac{p_{d1}}{\varphi_{d1}}\tilde{\omega}_{d1}^T\hat{\omega}_{d1} + \frac{k_{d1}}{\varphi_{d1}^*}\tilde{\zeta}_{d1}\hat{\zeta}_{d1} + \frac{3}{4r}. \end{aligned} \tag{41}$$

Moreover, with the help of Young's inequality, the following inequality can be obtained:

$$\varsigma_{d2}\dot{\varsigma}_{d2} \leq -\frac{\varsigma_{d2}^2}{\varpi_{d2}} + \frac{1}{2}\varsigma_{d2}^2 + \frac{1}{2}\dot{\alpha}_{d1}^2 \tag{42}$$

where

$$\begin{aligned}\dot{\alpha}_{d1} =\ & \frac{\partial\alpha_{d1}}{\partial\hat{x}_1}\dot{\hat{x}}_1 + \frac{\partial\alpha_{d1}}{\partial\hat{\zeta}_{d1}}\dot{\hat{\zeta}}_{d1} + \frac{\partial\alpha_{d1}}{\partial\hat{\omega}_{d1}}\dot{\hat{\omega}}_{d1} + \frac{\partial\alpha_{d1}}{\partial\hat{\epsilon}_{d1}}\dot{\hat{\epsilon}}_{d1}\\ & + \sum_{i=0}^{1}\frac{\partial\alpha_{d1}}{\partial y_r^i}y_r^{(i+1)}. \end{aligned} \tag{43}$$

Based on the derivation of the above equations, we choose $(1/\varpi_{d2}) \geq 1 + h_{d2}$ with $h_{d2} > 0$ being a design parameter. Immediately after that, obtain the following inequality:

$$\begin{aligned}\dot{\mathcal{V}}_1^d \leq\ & e^T\left(\mathcal{P}_d\mathcal{A}' + \mathcal{A}'^T\mathcal{P}_d + 6\mathcal{P}_d^T\mathcal{P}_d + \left(\frac{1}{k} + k\|\mathcal{P}_d\mathcal{L}_dM_d\|^2\right.\right.\\ & \left.\left. + \frac{3\vartheta_{dM}^2 + \vartheta_{dM}^4}{2}\right)I_n\right)e + 2\sum_{i=1}^{n}\tilde{\omega}_{di}^T\tilde{\omega}_{di} + q_{d1} + \frac{1}{2}(\hbar_2^d)^2\\ & - \sum_{i=1}^{n}\left(\vartheta_{di} - \frac{5}{2} - \vartheta_{dM}^2\right)\tilde{\lambda}_{di}^2 - \beta_{d1}(\hbar_1^d)^2 - h_{d2}\varsigma_{d2}^2\\ & + \tilde{\omega}_{d1}^T\tilde{\omega}_{d1} + \frac{p_{d1}}{\varphi_{d1}}\tilde{\omega}_{d1}^T\hat{\omega}_{d1} + \frac{k_{d1}}{\varphi_{d1}^*}\tilde{\zeta}_{d1}\hat{\zeta}_{d1} + \frac{1}{2}\dot{\alpha}_{d1}^2 \end{aligned} \tag{44}$$

where $q_{d1} = q_{d0} + \frac{3}{4r}$.

Step j ($2 \leq j \leq n-1$)**:** According to (31), the following equation holds:

$$\begin{aligned}\dot{\hbar}_j^d =\ & \hbar_{j+1}^d + \varsigma_{d,j+1} + \alpha_{dj} + \omega_{dj}^T\mathcal{S}_j^d(\hat{\bar{x}}_j) - \tilde{\omega}_{dj}^T\mathcal{S}_j^d(\hat{\bar{x}}_j)\\ & + \hat{\epsilon}_{dj} - \ell_j^d(\check{y}-\hat{x}_1) - \dot{\nu}_{dj}. \end{aligned} \tag{45}$$

The following Lyapunov function $\mathcal{V}_j^d$ is designed: $\mathcal{V}_j^d = \mathcal{V}_{j-1}^d + \frac{1}{2}(\hbar_j^d)^2 + \frac{1}{2\varphi_{dj}^*}\tilde{\zeta}_{dj}^2 + \frac{1}{2\varphi_{dj}}\tilde{\omega}_{dj}^T\tilde{\omega}_{dj} + \frac{1}{2}\varsigma_{d,j+1}^2$, where $\varphi_{dj} > 0$ and $\varphi_{dj}^* > 0$ are known constants. In addition, $\zeta_{dj} = \|\omega_{dj}\|^2$, $\hat{\zeta}_{dj}$ is the estimate of ζ_{dj} and $\tilde{\zeta}_{dj} = \zeta_{dj} - \hat{\zeta}_{dj}$.

Then, differentiating $\mathcal{V}_j^d$ shows that

$$\begin{aligned}\dot{\mathcal{V}}_j^d =\ & \dot{\mathcal{V}}_{j-1}^d + \hbar_j^d\left(\hbar_{j+1}^d + \varsigma_{d,j+1} + \alpha_{dj} + \omega_{dj}^T\mathcal{S}_j^d(\hat{\bar{x}}_j) + \hat{\epsilon}_{dj}\right.\\ & \left. -\tilde{\omega}_{dj}^T\mathcal{S}_j^d(\hat{\bar{x}}_j) - \ell_j^d(\check{y}-\hat{x}_1) - \dot{\nu}_{dj}\right) + \frac{1}{\varphi_{dj}}\tilde{\omega}_{dj}^T\dot{\hat{\omega}}_{dj}\\ & + \frac{1}{\varphi_{dj}^*}\tilde{\zeta}_{dj}\dot{\hat{\zeta}}_{dj} + \varsigma_{d,j+1}\dot{\varsigma}_{d,j+1}. \end{aligned} \tag{46}$$

Similar to the derivation process in the *Step 1*, the virtual controller α_{dj} is chosen as

$$\begin{aligned}\alpha_{dj} =\ & -\beta_{dj}\hbar_{dj} + \ell_j^d(\check{y}-\hat{x}_1) - \frac{5}{2}\hbar_{dj} + \dot{\nu}_{dj}\\ & - \frac{r}{2}\hbar_j^d\hat{\zeta}_{dj} - \hat{\epsilon}_{dj} \end{aligned} \tag{47}$$

and the adaptive laws are designed as

$$\begin{aligned}\dot{\hat{\omega}}_{dj} &= \varphi_{dj}\hbar_j^d\mathcal{S}_j^d(\hat{\bar{x}}_j) - p_{dj}\hat{\omega}_{dj}\\ \dot{\hat{\zeta}}_{dj} &= \varphi_{dj}^*\frac{r}{2}(\hbar_j^d)^2 - k_{dj}\hat{\zeta}_{dj}\end{aligned} \tag{48}$$

where $\beta_{dj} > 0$, $p_{dj} > 0$ and $k_{dj} > 0$ are know constants.

Based on Young's inequality, we have

$$\varsigma_{d,j+1}\dot{\varsigma}_{d,j+1} \leq -\frac{\varsigma_{d,j+1}^2}{\varpi_{d,j+1}} + \frac{1}{2}\varsigma_{d,j+1}^2 + \frac{1}{2}\dot{\alpha}_{dj}^2 \tag{49}$$

where

$$\begin{aligned}\dot{\alpha}_{dj} =\ & \sum_{i=1}^{j}\frac{\partial\alpha_{dj}}{\partial\hat{x}_i}\dot{\hat{x}}_i + \sum_{i=1}^{j}\frac{\partial\alpha_{dj}}{\partial\hat{\zeta}_{di}}\dot{\hat{\zeta}}_{di} + \sum_{i=1}^{j}\frac{\partial\alpha_{dj}}{\partial\hat{\omega}_{di}}\dot{\hat{\omega}}_{di}\\ & + \sum_{i=1}^{j}\frac{\partial\alpha_{dj}}{\partial\hat{\epsilon}_{di}}\dot{\hat{\epsilon}}_{di} + \sum_{i=0}^{j}\frac{\partial\alpha_{dj}}{\partial y_r^i}y_r^{(i+1)}. \end{aligned} \tag{50}$$

Then, we choose $(1/\varpi_{d,j+1}) \geq 1 + h_{d,j+1}$ with $h_{d,j+1} > 0$ being the design parameter. Thus, one obtains that

$$\begin{aligned}\dot{\mathcal{V}}_j^d \leq\ & e^T\left(\mathcal{P}_d\mathcal{A}' + \mathcal{A}'^T\mathcal{P}_d + 6\mathcal{P}_d^T\mathcal{P}_d + \left(\frac{1}{k} + k\|\mathcal{P}_d\mathcal{L}_dM_d\|^2\right.\right.\\ & \left.\left. + \frac{3\vartheta_{dM}^2 + \vartheta_{dM}^4}{2}\right)I_n\right)e + 2\sum_{i=1}^{n}\tilde{\omega}_{di}^T\tilde{\omega}_{di} + \frac{1}{2}\sum_{i=1}^{j}\dot{\alpha}_{di}^2\\ & - \sum_{i=1}^{n}\left(\vartheta_{di} - \frac{5}{2} - \vartheta_{dM}^2\right)\tilde{\lambda}_{di}^2 - \sum_{i=1}^{j}\beta_{di}(\hbar_i^d)^2\\ & - \sum_{i=2}^{j+1}h_{di}\varsigma_{di}^2 + \sum_{i=1}^{j}\tilde{\omega}_{di}^T\tilde{\omega}_{di} + \frac{1}{2}(\hbar_{j+1}^d)^2\\ & + \sum_{i=1}^{j}\frac{p_{di}}{\varphi_{di}}\tilde{\omega}_{di}^T\hat{\omega}_{di} + \sum_{i=1}^{j}\frac{k_{di}}{\varphi_{di}^*}\tilde{\zeta}_{di}\hat{\zeta}_{di} + q_{dj} \end{aligned} \tag{51}$$

where $q_{dj} = q_{d,j-1} + \frac{1}{2r}$.

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Step n: Based on the coordinate transformation defined in (31), we can know that the derivative of $\hbar_{dn}$ is

$$\dot{\hbar}_{dn} = u_d + \omega_{dj}^T \mathcal{S}_j^d(\hat{\bar{x}}_n) - \tilde{\omega}_{dn}^T \mathcal{S}_n^d(\hat{\bar{x}}_n) + \hat{\epsilon}_{dn} - \ell_n^d(\check{y} - \hat{x}_1) - \dot{\nu}_{dn} + u_d^* - u_d^*. \tag{52}$$

The Lyapunov function $\mathcal{V}_n^d$ is chosen as $\mathcal{V}_n^d = \mathcal{V}_{n-1}^d + \frac{1}{2}(\hbar_n^d)^2 + \frac{1}{2\varphi_{dn}^*}\tilde{\zeta}_{dn}^2 + \frac{1}{2\varphi_{dn}}\tilde{\omega}_{dn}^T\tilde{\omega}_{dn}$, where $\varphi_{dn} > 0$ and $\varphi_{dn}^* > 0$ are known constants. In addition, $\zeta_{dn} = ||\omega_{dn}||^2$, $\hat{\zeta}_{dn}$ is the estimate of ζ_{dn} and $\tilde{\zeta}_{dn} = \zeta_{dn} - \hat{\zeta}_{dn}$.

Then, we have

$$\dot{\mathcal{V}}_n^d = \dot{\mathcal{V}}_{n-1}^d + \hbar_n^d \left(u_d + \omega_{dn}^T \mathcal{S}_n^d(\hat{\bar{x}}_n) + \hat{\epsilon}_{dn} - \tilde{\omega}_{dn}^T \mathcal{S}_n^d(\hat{\bar{x}}_n) - \ell_n^d(\check{y} - \hat{x}_1) - \dot{\nu}_{dn} + u_d^* - u_d^*\right) + \frac{1}{\varphi_{dn}}\tilde{\omega}_{dn}^T\dot{\tilde{\omega}}_{dn} + \frac{1}{\varphi_{dn}^*}\tilde{\zeta}_{dn}\dot{\tilde{\zeta}}_{dn}. \tag{53}$$

By introducing Young's inequality, the following inequalities always holds:

$$\begin{aligned} \hbar_n^d(u_d - u_d^*) &\le \frac{1}{4}(\hbar_n^d)^2 + \Delta_d^2, \\ \hbar_n^d \omega_{dn}^T \mathcal{S}_n^d(\hat{\bar{x}}_n) &\le \frac{r}{2}(\hbar_n^d)^2\zeta_{dn} + \frac{1}{2r}, \\ -2\hbar_n^d\tilde{\omega}_{dn}^T\mathcal{S}_n^d(\hat{\bar{x}}_n) &\le (\hbar_n^d)^2 + \tilde{\omega}_{dn}^T\tilde{\omega}_{dn} \end{aligned} \tag{54}$$

where r is a known positive number.

In conjunction with (53) and (54), we obtain

$$\dot{\mathcal{V}}_n^d \le \dot{\mathcal{V}}_{n-1}^d + \hbar_n^d\Big(u_d^* + \frac{r}{2}\hbar_n^d\hat{\zeta}_{dn} + \hat{\epsilon}_{dn} - \dot{\nu}_{dn} + \frac{5}{4}(\hbar_n^d)^2 - \ell_n^d(\check{y} - \hat{x}_1)\Big) + \tilde{\omega}_{dn}^T\tilde{\omega}_{dn} + \frac{1}{2r} + \Delta_d^2 + \tilde{\omega}_{dn}^T\left(\hbar_n^d\mathcal{S}_n^d(\hat{\bar{x}}_n) - \frac{1}{\varphi_{dn}}\dot{\hat{\omega}}_{dn}\right) + \tilde{\zeta}_{dn}\left(\frac{r}{2}(\hbar_n^d)^2 - \frac{1}{\varphi_{dn}^*}\dot{\hat{\zeta}}_{dn}\right). \tag{55}$$

The control command u_d^* is designed as follows:

$$u_d^* = -\beta_{dn}\hbar_n^d + \ell_n^d(\check{y} - \hat{x}_1) - \frac{7}{4}\hbar_n^d + \dot{\nu}_{dn} - \frac{r}{2}\hbar_n^d\hat{\zeta}_{dn} - \hat{\epsilon}_{dn} \tag{56}$$

and the adaptive laws are selected as

$$\begin{aligned} \dot{\hat{\omega}}_{dn} &= \varphi_{dn}\hbar_n^d\mathcal{S}_n^d(\hat{\bar{x}}_n) - p_{dn}\hat{\omega}_{dn} \\ \dot{\hat{\zeta}}_{dn} &= \varphi_{dn}^*\frac{r}{2}(\hbar_n^d)^2 - k_{dn}\hat{\zeta}_{dn} \end{aligned} \tag{57}$$

where $\beta_{dn} > 0$, $p_{dn} > 0$, and $k_{dn} > 0$ are know constants.

Based on the above, we have

$$\dot{\mathcal{V}}_n^d \le e^T\left(\mathcal{P}_d\mathcal{A}' + \mathcal{A}'^T\mathcal{P}_d + 6\mathcal{P}_d^T\mathcal{P}_d + \left(\frac{1}{k} + k\|\mathcal{P}_d\mathcal{L}_dM_d\|^2 + \frac{3\vartheta_{dM}^2 + \vartheta_{dM}^4}{2}\right)I_n\right)e + 3\sum_{i=1}^n\tilde{\omega}_{di}^T\tilde{\omega}_{di} + \frac{1}{2}\sum_{i=1}^{n-1}\dot{\alpha}_{di}^2 - \sum_{i=1}^n\left(\vartheta_{di} - \frac{5}{2} - \vartheta_{dM}^2\right)\tilde{\lambda}_{di}^2 - \sum_{i=1}^n\beta_{di}(\hbar_i^d)^2 - \sum_{i=2}^n h_{di}\varsigma_{di}^2 + \sum_{i=1}^n\frac{p_{di}}{\varphi_{di}}\tilde{\omega}_{di}^T\hat{\omega}_{di} + \sum_{i=1}^n\frac{k_{di}}{\varphi_{di}^*}\tilde{\zeta}_{di}\hat{\zeta}_{di} + q_{dn} \tag{58}$$

where $q_{dn} = q_{d,n-1} + \frac{1}{2r} + \Delta_d^2$.

The following inequalities can be obtained:

$$\begin{aligned} \tilde{\omega}_{di}^T\hat{\omega}_{di} &\le -\frac{1}{2}\tilde{\omega}_{di}^T\tilde{\omega}_{di} + \frac{1}{2}\hat{\omega}_{di}^T\hat{\omega}_{di}, \\ \tilde{\zeta}_{di}\hat{\zeta}_{di} &\le -\frac{1}{2}\tilde{\zeta}_{di}^2 + \frac{1}{2}\hat{\zeta}_{di}^2. \end{aligned} \tag{59}$$

For $\forall\ \Xi_d > 0$, the set $\Upsilon_d := \{\mathcal{V}^d | \mathcal{V}^d = 2e^T\mathcal{P}_de + \sum_{i=1}^n\tilde{\lambda}_{di}^2 + \sum_{i=1}^n(\hbar_i^d)^2 + \sum_{i=1}^n\frac{1}{\varphi_{di}^*}\tilde{\zeta}_{di}^2 + \sum_{i=1}^n\frac{1}{\varphi_{di}}\tilde{\omega}_{di}^T\tilde{\omega}_{di} + \sum_{i=2}^{n-1}\varsigma_{di}^2 \le 2\Xi_d\}$ is compact in R^{2n+2}. Thus, $|\dot{\alpha}_j^a| \le \aleph_d$ on Ξ_d for $j = 1, \ldots, n-1$. Then, the following inequality holds:

$$\dot{\mathcal{V}}^d \le e^T\left(\mathcal{P}_d\mathcal{A}' + \mathcal{A}'^T\mathcal{P}_d + 6\mathcal{P}_d^T\mathcal{P}_d + \left(\frac{1}{k} + k\|\mathcal{P}_d\mathcal{L}_dM_d\|^2 + \frac{3\vartheta_{dM}^2 + \vartheta_{dM}^4}{2}\right)I_n\right)e + 3\sum_{i=1}^n\tilde{\omega}_{di}^T\tilde{\omega}_{di} - \sum_{i=1}^n\beta_{di}(\hbar_i^d)^2 - \sum_{i=1}^n\left(\vartheta_{di} - \frac{5}{2} - \vartheta_{dM}^2\right)\tilde{\lambda}_{di}^2 - \sum_{i=2}^n h_{di}\varsigma_{di}^2 + q_d - \sum_{i=1}^n\frac{p_{di}}{2\varphi_{di}}\tilde{\omega}_{di}^T\tilde{\omega}_{di} - \sum_{i=1}^n\frac{k_{di}}{2\varphi_{di}^*}\tilde{\zeta}_{di}^2 \tag{60}$$

where $q_d = q_{dn} + (n/2)\aleph_d^2 + \sum_{i=1}^n\frac{p_{di}}{2\varphi_{di}}\hat{\omega}_{di}^T\hat{\omega}_{di} + \sum_{i=1}^n\frac{k_{di}}{2\varphi_{di}^*}\hat{\zeta}_{di}^2$.

Case 2 : When $t \in [T_l^s, T_{l+1}^d)$, the DoS attack is sleeping. In this case, the tracking error $z_{s,1} = y - y_r$.

The remainder of the derivation is similar to *Case 1*, and the final control command u_s^* can be designed to be

$$u_s^* = -\beta_{sn}\hbar_n^s + \ell_n^s(\check{y} - \hat{x}_1) - \frac{7}{4}\hbar_n^s + \dot{\nu}_{sn} - \frac{r}{2}\hbar_n^s\hat{\zeta}_{sn} - \hat{\epsilon}_{sn} \tag{61}$$

and the adaptive laws are selected as

$$\begin{aligned} \dot{\hat{\omega}}_{sn} &= \varphi_{sn}\hbar_n^s\mathcal{S}_n^s(\hat{\bar{x}}_n) - p_{sn}\hat{\omega}_{sn} \\ \dot{\hat{\zeta}}_{sn} &= \varphi_{sn}^*\frac{r}{2}(\hbar_n^s)^2 - k_{sn}\hat{\zeta}_{sn} \end{aligned} \tag{62}$$

where $\beta_{sn} > 0$, $p_{sn} > 0$, and $k_{sn} > 0$ are known constants.

Thus, we can finally get the following inequality:

$$\dot{\mathcal{V}}^s \le e^T\left(\mathcal{P}_s\mathcal{A} + \mathcal{A}^T\mathcal{P}_s + 6\mathcal{P}_s^T\mathcal{P}_s + \frac{3\vartheta_{sM}^2 + \vartheta_{sM}^4 + 2}{2}I_n\right)e + 2\sum_{i=1}^n\tilde{\omega}_{si}^T\tilde{\omega}_{si} + q_s + \sum_{i=2}^n\tilde{\omega}_{si}^T\tilde{\omega}_{si} - \sum_{i=1}^n\beta_{si}(\hbar_i^s)^2 - \sum_{i=1}^n(\vartheta_{si} - 3 - \vartheta_{sM}^2)\tilde{\lambda}_{si}^2 - \sum_{i=2}^n h_{si}\varsigma_{si}^2 - \sum_{i=1}^n\frac{p_{si}}{2\varphi_{si}}\tilde{\omega}_{si}^T\tilde{\omega}_{si} - \sum_{i=1}^n\frac{k_{si}}{2\varphi_{si}^*}\tilde{\zeta}_{si}^2 \tag{63}$$

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where $q_s = q_{sn} + (n/2)\aleph_s^2 + \sum_{i=1}^{n} \frac{p_{si}}{2\varphi_{si}} \hat{\omega}_{si}^T \hat{\omega}_{si} + \sum_{i=1}^{n} \frac{k_{si}}{2\varphi_{si}^*} \hat{\zeta}_{si}^2$.

In order to ensure the stability of the error system (13), the observer design parameters $\mathcal{L}_d$ and $\mathcal{L}_s$ are chosen to satisfy the following conditions:

$$\mathcal{P}_d \mathcal{A}' + \mathcal{A}'^T \mathcal{P}_d + 6\mathcal{P}_d^T \mathcal{P}_d + \left(\frac{1}{k} + k\|\mathcal{P}_d \mathcal{L}_d M_d\|^2 + \frac{3\vartheta_{dM}^2 + \vartheta_{dM}^4}{2} \right) I_n \le \Pi_d \mathcal{P}_d \tag{64}$$

and

$$\mathcal{P}_s \mathcal{A} + \mathcal{A}^T \mathcal{P}_s + 6\mathcal{P}_s^T \mathcal{P}_s + \frac{3\vartheta_{sM}^2 + \vartheta_{sM}^4 + 2}{2} I_n \le -\Pi_s \mathcal{P}_s \tag{65}$$

where Π_d and Π_s are positive constants.

Furthermore, (60) and (62) can be rewritten as

$$\begin{aligned} \dot{\mathcal{V}}^d \le\ & \Pi_d e^T \mathcal{P}_d e + q_d - \sum_{i=1}^{n} \beta_{di} (\hbar_i^d)^2 - \sum_{i=2}^{n} h_{di} \varsigma_{di}^2 \\ & - \sum_{i=1}^{n} \left(\vartheta_{di} - \frac{5}{2} - \vartheta_{dM}^2 \right) \tilde{\lambda}_{di}^2 - \sum_{i=1}^{n} \frac{k_{di}}{2\varphi_{di}^*} \tilde{\zeta}_{di}^2 \\ & - \sum_{i=1}^{n} \left(\frac{p_{di}}{2\varphi_{di}} - 3 \right) \tilde{\omega}_{di}^T \tilde{\omega}_{di} \end{aligned} \tag{66}$$

and

$$\begin{aligned} \dot{\mathcal{V}}^s \le\ & -\Pi_s e^T \mathcal{P}_s e + q_s - \sum_{i=1}^{n} (\vartheta_{si} - 3 - \vartheta_{sM}^2) \tilde{\lambda}_{si}^2 \\ & - \sum_{i=1}^{n} \beta_{si} (\hbar_i^s)^2 - \tilde{\omega}_{s1}^T \tilde{\omega}_{s1} - \sum_{i=2}^{n} h_{si} \varsigma_{si}^2 \\ & - \sum_{i=1}^{n} \left(\frac{p_{si}}{2\varphi_{si}} - 3 \right) \tilde{\omega}_{si}^T \tilde{\omega}_{si} - \sum_{i=1}^{n} \frac{k_{si}}{2\varphi_{si}^*} \tilde{\zeta}_{si}^2. \end{aligned} \tag{67}$$

Meanwhile, the design parameters β_{dj}, β_{sj}, h_{dj}, h_{sj}, p_{dj}, p_{sj}, k_{di}, k_{si}, ϑ_{dj}, ϑ_{sj}, φ_{dj}, φ_{sj}, φ_{dj}^*, and $\varphi_{sj}^* (j = 1, \ldots, n)$ are selected by $-2\beta_{dj} < \Pi_d$, $-2h_{dj} < \Pi_d$, $-2(\frac{p_{di}}{2\varphi_{di}} - 3) < \Pi_d$, $-\frac{k_{di}}{\varphi_{di}^*} < \Pi_d$, $-2(\vartheta_{di} - \frac{5}{2} - \vartheta_{dM}^2) < \Pi_d$ and $-2\beta_{sj} \le -\Pi_s$, $-2h_{sj} \le -\Pi_s$, $-2(\frac{p_{si}}{2\varphi_{si}} - 3) \le -\Pi_s$, $-\frac{k_{si}}{\varphi_{si}^*} \le -\Pi_s$, $-2(\vartheta_{si} - 3 - \vartheta_{sM}^2) \le -\Pi_s$.

According to (66) and (67), the following inequalities hold:

$$\dot{\mathcal{V}}^d \le \Pi_d \mathcal{V}^d + q_d, \quad \dot{\mathcal{V}}^s \le -\Pi_s \mathcal{V}^s + q_s. \tag{68}$$

In turn, the derivative of the final Lyapunov function can be expressed as

$$\dot{\mathcal{V}} = \begin{cases} \dot{\mathcal{V}}^d, & t \in [T_l^d, T_l^s), \\ \dot{\mathcal{V}}^s, & t \in [T_l^s, T_{l+1}^d). \end{cases} \tag{69}$$

C. System Stability

Theorem 1: Consider the nonlinear system (1) under DoS attacks that satisfies Assumptions 1–4, as well as Lemmas 1–2, the DETM (9), the composite observer (12), the parameter laws (40), (42), (48), (57), and (62). In addition, if there exist positive definite matrices $\mathcal{P}_d, \mathcal{P}_s$, the vectors $\mathcal{L}_d, \mathcal{L}_s$ and constant ξ, they are satisfied

$$\mathcal{P}_d < \sigma \mathcal{P}_d, \mathcal{P}_s < \sigma \mathcal{P}_s, \quad \frac{\Pi_s - \xi}{\Pi_s + \Pi_d} > \frac{1}{\mathcal{T}}, \xi > \frac{\ln \theta}{\tau_D}. \tag{70}$$

Then, the designed adaptive fuzzy safety control strategy can guarantee that all signals of the closed-loop system are bounded. Besides, the Zeno behavior caused by the DETM can be avoided.

Proof: Under the conditions (68)–(70), the following inequality holds: $\dot{\mathcal{V}} \le \delta(t)\mathcal{V} + q$, where $\delta(t) = \Pi_d$ if $t \in [T_l^d, T_l^s)$, $\delta(t) = -\Pi_s$ if $t \in [T_l^s, T_{l+1}^d)$, and $q = q_d + q_s$. Now, for $t \in [T_l^d, T_l^s)$ to be considered by us, the following inequality holds:

$$\begin{aligned} \mathcal{V}(t) &\le e^{\delta_l (t - T_l^d)} \mathcal{V}(T_l^{d+}) + q \int_{T_l^d}^{t} e^{\delta_l (t-\varepsilon)} d\varepsilon \\ &\le \theta \mathcal{V}(T_l^{d-}) + q \int_{T_l^d}^{t} e^{\delta_l (t-\varepsilon)} d\varepsilon. \end{aligned} \tag{71}$$

In the lth DoS attack interval, a recursive approach can be employed to obtain

$$\begin{aligned} \mathcal{V}(t) \le\ & \theta^{n(0,t)} e^{\delta_l (t - T_l^d) + \cdots + \delta_0 (T_1^d - 0)} \mathcal{V}(0) \\ & + q \int_0^{T_1^d} \theta^{n(T_1^d, t)} e^{\delta_l (t - T_l^d) + \cdots + \delta_0 (T_1^d - \varepsilon)} d\varepsilon \\ & + \cdots + q \int_{T_l^d}^{t} e^{\varrho_p (t-\varepsilon)} d\varepsilon. \end{aligned} \tag{72}$$

Under Assumption 3, we have $|\Psi_d(\varepsilon, t)| \le (t - \varepsilon/\mathcal{T})$ and $|\Psi_s(\varepsilon, t)| \ge (t - \varepsilon)(1 - (1/\mathcal{T}))$ on the time interval $[\varepsilon, t]$. Further, the following inequality can be obtained:

$$\begin{aligned} \mathcal{V}(t) \le\ & \theta^{n(0,t)} e^{-[\Pi_s |\Psi_s(0,t)| - \Pi_d |\Psi_s(0,t)|]} \mathcal{V}(0) \\ & + q \int_0^{T_1^d} \theta^{n(T_1^d, t)} e^{-[\Pi_s |\Psi_s(\varepsilon,t)| - \Pi_d |\Psi_d(\varepsilon,t)|]} d\varepsilon \\ \le\ & e^{-[\Pi_s |\Psi_s(0,t)| - \Pi_d |\Psi_d(0,t)| - n(0,t) \ln \theta]} \mathcal{V}(0) \\ & + q \int_0^{t} e^{-[\Pi_s |\Psi_s(0,t)| - \Pi_d |\Psi_d(0,t)| - n(\varepsilon,t) \ln \theta]} d\varepsilon. \end{aligned} \tag{73}$$

In the light of Assumption 4 and (70), there exists a constant $\phi > 0$ satisfies $\Pi_s |\Psi_s(\varepsilon, t)| - \Pi_d |\Psi_d(\varepsilon, t)| > \xi(t - \varepsilon) > (n(\varepsilon, t) - \eta) \ln \theta + \phi(t - \varepsilon)$.

Thus, we have

$$\begin{aligned} \mathcal{V}(t) &\le \theta^{\eta} \left[e^{-\phi t} \mathcal{V}(0) + q \int_0^t e^{-\phi(t-\varepsilon)} d\varepsilon \right] \\ &\le \theta^{\eta} \left[e^{-\phi t} \mathcal{V}(0) + \frac{q}{\phi} \right]. \end{aligned} \tag{74}$$

Besides, it is easy to obtain that $\dot{\mathcal{V}} \le -\phi \mathcal{V}(t) + q$. Then, this implies that the following inequality holds:

$$\mathcal{V}(t) \le e^{-\phi t} \mathcal{V}(0) + \frac{q}{\phi} (1 - e^{-\phi t}). \tag{75}$$

Based on the above derivations, we can deduce that $\mathcal{V} \in \mathbb{L}_\infty$, which implies that all signals in the closed-loop system are bounded. Because $\mathfrak{Z}_1$ is bounded, we get $-g(t) < z_1 < g(t)$ holds, and then by definition of $g(t)$, we finally get $|z_1| < b$ for $t > T$.

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In order to simplify the analysis process, we use u and u^* to denote the real controller and the final control command, respectively. Note that during the interval $t \in [t_k, t_{k+1})$, e_u starts at zero and may hit zero again several times in the subsequent process. Let t_I be the last time instant that $|e_u|$ hits zero, then $t_k \leq t_I < t_{k+1}$ holds. Moreover, there also exist $|e_u| > 0$ in the interval (t_I, t_{k+1}). Combining (9), we can obtain that the controller u remains constant for $t \in [t_k, t_{k+1})$, which implies that in this time period $\dot{u} = 0$. Immediately afterwards, the derivative of e_u in the interval $[t_I, t_{k+1})$ is $\frac{d|e_u|}{dt} \leq |\dot{u} - \dot{u}^*| = |\dot{u}^*|$ and thus the following equation holds:

$$\dot{u}^* = \frac{\partial u^*}{\partial \hbar_n}\dot{\hbar}_n + \frac{\partial u^*}{\partial \hat{\zeta}_n}\dot{\hat{\zeta}}_n + \frac{\partial u^*}{\partial \hat{\epsilon}_n}\dot{\hat{\epsilon}}_n. \tag{76}$$

Based on the above analysis, it can be obtained that $\hbar_n$, $\hat{\zeta}_n$ and $\hat{\epsilon}_n$ are continuously differentiable and bounded. In turn, $\dot{u}^*$ are bounded. Therefore, there exists a positive constant $\bar{u}^*$ such that $|\dot{u}^*| \leq |\bar{u}^*|$ holds. Considering $e_u(t_I) = 0$ and $\lim_{t \to t_{k+1}} |e_u| = \Delta$, we can get the lower bound of the interexecution interval $\mathcal{I}$ to satisfy the following condition: $\mathcal{I} = t_{k+1} - t_I + t_I - t_k \geq \frac{\Delta}{\bar{u}^*} + t_I - t_k$. This means that the phenomenon of infinite triggering will not occur, that is, the behavior of Zeno is avoided. Therefore, the proof of Theorem 1 ends.

IV. Simulation Studies

This section verifies the effectiveness and rationality of the designed dynamic event-triggered adaptive prescribed-time safety control scheme based on composite observer.

Consider the single-link manipulator [33] subjected to external disturbances, the mathematical model is shown below

$$B\ddot{v} + Q\dot{v} + \frac{1}{2}MgL\sin(v) = u(t) + \epsilon(t) \tag{77}$$

where $x_1 = v$ is the angle of the link, $x_2 = \dot{v}$ is the angular velocity, B is the total rotational inertia of the motor, Q is the total damping coefficient, L is the distance from the joint axis to the center of mass, g is the acceleration of gravity, and M is the total mass of the link. Furthermore, we define the ideal trajectory $y_r = \sin(0.5t)$.

In the simulation, the desired tracking accuracy $b = 3.5$ and the predefined time $\mathcal{T} = 1$ s. Let $B = 1$ kgm^2, $Q = 2$ kgm/s, $g = 9.8$ m/s^2, $L = 1$ m, $M = 1$ kg. The initial states are selected as $x_1(0) = 0.1$ $x_2(0) = 0.2$ and other relevant parameters are selected as $r = 2$, $\vartheta_1^d = \vartheta_1^s = \vartheta_2^d = \vartheta_2^s = 0.1$, $\varpi_2^d = \varpi_2^s = 0.5$, $\beta_1^d = \beta_2^d = 20$, $\beta_1^s = \beta_2^s = 50$, $\mathcal{L}_d = [-1.159, -10.076]^T$, $\mathcal{L}_s = [-10.021, -100.171]^T$, $r_u = 1$, $o = 0.8$, $\hbar = 0.6$. The intervals in which the attack occurs can be chosen as $[10.62\text{ s}, 12.6\text{ s}] \cup [19.62\text{ s}, 23.6\text{ s}] \cup [30.62\text{ s}, 33.6\text{ s}] \cup [42.62\text{ s}, 45.6\text{ s}]$.

The simulation results for the single-link robotic arm subjected to external disturbances are shown in Figs. 1–2. Fig. 1(a) depicts the tracking error. The result shows that $|z_1(t)| < b$ always holds at $t \geq \mathcal{T}$. The errors of the composite observer are shown in Fig. 1(b). Moreover, Fig. 2 represents the triggering time of the system input, indicating that the execution interval is strictly positive.

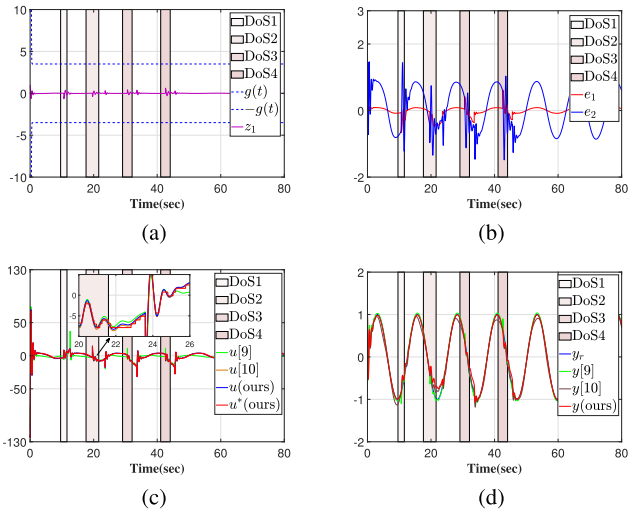


Fig. 1. (a) Tracking error z_1, (b) estimation errors e_1 and e_2, (c) input u under different strategies, (d) tracking trajectories under different strategies.

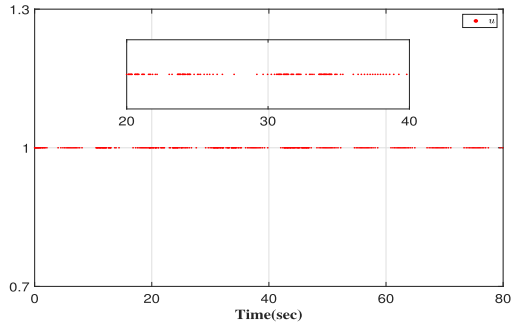


Fig. 2. Triggering times of the control input.

In order to be able to show the effectiveness of the proposed control method, we compare it with the schemes in [9] and [10]. Note that the methods proposed in [9] and [10] do not involve unknown disturbances. Therefore, we consider using the following single-link manipulator $B\ddot{v} + Q\dot{v} + \frac{1}{2}MgL\sin(v) = u(t)$, which is not subject to external disturbances, in order to make a fair comparison. In addition, we use the same parameters as in the article case. The simulation results are displayed in Fig. 1(c) and (d). Compared to the existing methods in [9] and [10], our proposed control strategy has the following advantages: 1) it can better accomplish the tracking of the desired trajectory under DoS attacks; 2) our proposed control method can save more communication resources; and 3) in the presence of unknown disturbances, the controller is still able to guarantee a similar performance as in [9] and [10].

V. Conclusion

This article addressed the problem of dynamic event-triggered adaptive prescribed-time fuzzy control for nonlinear systems with external disturbances under DoS attacks. A new composite observer was designed using the compensated output signal of the attack compensator to achieve the reconstructions of system

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state, as well as the estimates of the unknown disturbances. By combining the time-varying constraint function with the backstepping method, the safety fuzzy controller was built to ensure that the tracking was achieved within the prescribed time. In addition, based on the introduced dynamic auxiliary variable, we designed the DETM within the channel from the controller to the actuator, which saves communication resources. Finally, the simulation results verify the rationality and effectiveness of this control strategy. Our future research direction is to extend this control strategy to the nonlinear heterogeneous MASs.

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